Solutions and teaching notes for chapter 5

Teacher use. The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $\frac{12}{40} = \frac{3}{10} = 30\%$. *Action:* Link the partial count to the reference count.
2. 0.7. *Action:* Represent the shared sample space as two events.
3. $A \cap B$. *Action:* Shade the intersection of two sets.
4. $A \cup B$. *Action:* Include the common outcomes in the union.
5. 0.1. *Action:* Replace the decimals by fractions.
6. $\frac{3}{6} = \frac{1}{2}$. *Action:* Identify the equally-likely assumption.

Investigation activity

Required intersection 12; cells 12, 28, 18, 42. Replacing them by 13, 27, 17, 43 keeps the totals but $\frac{13}{100} \neq 0.4 \times 0.3$.

Immediate applications

1. $\frac{5}{60} = \frac{1}{12}$ and $\frac{5}{20} = \frac{1}{4}$.
2. $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$.
3. No: $0.4 \times 0.5 = 0.20 \neq 0.18$.
4. $(\frac{2}{3})^4 = \frac{16}{81}$ then $1 - \frac{16}{81} = \frac{65}{81}$.
5. $\frac{675}{2000} = 0.3375$; this is an estimate, not the exact value.

Exercises: method and results

Exercise 5.1 – A table. Cells: 15 (A and B), 25 (A without B), 20 (B without A), 40 (neither A nor B). $P_A(B) = \frac{15}{40} = \frac{3}{8}$; $P_B(A) = \frac{15}{35} = \frac{3}{7}$.

Exercise 5.2 – Production. $P(D) = 0.25 \times 0.04 + 0.75 \times 0.02 = 0.025$. $P_D(A) = \frac{0.01}{0.025} = 0.4$. The denominator changes when we condition on D.

Exercise 5.3 – Checking independence. $P(A) = P(B) = \frac{1}{2}$; the intersection $\{4,6\}$ has probability $\frac{1}{3} \neq \frac{1}{4}$: not independent. Unlike the example in the course (result at least 5), here the threshold 4 does not give independence.

Exercise 5.4 – Without replacement. Two red: $(\frac{3}{5})(\frac{2}{4}) = \frac{3}{10}$. One of each: $(\frac{3}{5})(\frac{2}{4}) + (\frac{2}{5})(\frac{3}{4}) = \frac{3}{5}$. The order distinguishes two disjoint paths.

Exercise 5.5 – Three trials. One success: three paths, each $(\frac{1}{3})(\frac{2}{3})^2 = \frac{4}{27}$, total $\frac{4}{9}$. Two successes: $3(\frac{1}{3})^2(\frac{2}{3}) = \frac{2}{9}$; three: $\frac{1}{27}$; total $\frac{7}{27}$.

Exercise 5.6 – Detector. $P(S) = 0.02(0.9) + 0.98(0.05) = 0.067$. $P_S(D) = \frac{0.018}{0.067} = \frac{18}{67} \approx 26.9\%$. The low initial rate explains the gap with 90%.

Exercise 5.7 – Sets and cardinality. $A \cap B = \{2\}$ (1); $A \cup B = \{1,2,4\}$ (3); $E \setminus A = \{3,4\}$ (2); product $\{(1,2),(1,4),(2,2),(2,4)\}$ (4). A pair is ordered.

Exercise 5.8 – Delayed review. $P(B) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$. $P_B(A) = \frac{\frac{1}{8}}{\frac{1}{2}} = \frac{1}{4}$. The numerical equality with $P_A(B)$ is particular to these data.

CQFD, exercise 67 p. 336, questions 1, 2 and 3. $P(\overline{D} \cap \overline{N}) = 0.9 \times 0.05 = 0.045$. The probability of a positive test is $0.1 \times 0.97 + 0.045 = 0.142$. Among the positive tests, the probability of not being doped is $\frac{0.045}{0.142} = \frac{45}{142} \approx 31.7\%$. Do not swap positive and negative test.

CQFD, exercise 70 p. 336, questions a and b. $P(B) = \frac{1}{4}$, $P_A(B) = \frac{1}{2}$ and $P_{\overline{A}}(B) = 0$. Hence $\frac{1}{4} \neq \frac{1}{2} + 0$, then $0 \neq 1 - \frac{1}{2}$. The law of total probability involves the weights $P(A)$ and $P(\overline{A})$; the complement applies to the event B under the same condition: $P_A(B) + P_A(\overline{B}) = 1$, and not $P_A(B) + P_{\overline{A}}(B) = 1$.

MCQ: answers and reasons

MCQ 1.

1. B. Reference sample space A.
2. A. Product of the branches.
3. D. Definition.
4. C. Criterion without division.
5. B. Zero intersection, positive product.

MCQ 2.

1. C. Complement of three faces.
2. A. Product p^2.
3. D. Probabilities of a conditional partition.
4. B. Observed frequency and model probability differ.
5. C. Two red balls remain out of 4.

True or false

1. False: 12 common objects out of 40 A and 30 B give $\frac{3}{10}$ and $\frac{2}{5}$.
2. True: $P(A \cap \overline{B}) = P(A) - P(A \cap B) = P(A)(1 - P(B))$.
3. False: without replacement, the composition of the urn changes.

Differentiated teaching. For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.